

Name - Saikat Sheet
Batch - 25SUB5936_Linux System Programming
ID - 68500

Task 16:

Debugging with GDB

Write a C program that intentionally contains a logical error (e.g., incorrect loop condition).

Use GDB (GNU Debugger) to debug the program:

Set breakpoints, inspect variable values, and step through the program to find and fix the error.

Commands:

[Creating a buggy C Program]

> vi buggy.c

[This program is supposed to print numbers from 1 to 5, but because of a typo in the loop condition (`i > 5` instead of `i <= 5`), the loop never runs.]

`#include <stdio.h>`

```
int main() {
```

```
    int i = 1;
```

```
    // LOGICAL ERROR: The condition is upside down!
```

```
    while (i > 5) {
```

```
        printf("Number: %d\n", i);
```

```
        i++;
```

```
    }
```

```
    printf("Loop finished. Value of i is: %d\n", i);
```

```
    return 0;
```

```
}
```

[Compile for Debugging: To debug a program with GDB, we must compile it with the `-g` flag. This flag tells the compiler to include extra debugging details inside the executable.

```
> gcc -g buggy.c -o buggy
```

[Debugging with GDB]

```
> gdb ./buggy
```

[Set a Breakpoint]

```
> (gdb) break 6
```

[Run the Program]

```
> (gdb) run
```

[Inspect the Variable]

```
> (gdb) print i
```

[Step Through]

```
> (gdb) next
```

Output

Instead of jumping inside the loop, the program instantly jumps all the way down to line 11: `printf("Loop finished...")`.

Because we printed `i` and saw it was a wrong number, we can instantly realize why `while (1 > 5)` failed and skipped the loop entirely.

```
student@student: ~/25SUB5936_U21/Assignments/Day32
student@student:~$ cd 25SUB5936_U21/Assignments/Day32
student@student:~/25SUB5936_U21/Assignments/Day32$ ll
total 476
drwxrwxr-x  2 student student 4096 May 14 21:06 ./
drwxrwxr-x 37 student student 4096 May 17 11:26 ../
-rw-rw-r--  1 student student 745 May 14 21:05 hello_syscall.c
-rw-rw-r--  1 student student 715 May 14 21:05 reverse_str.c
-rw-rw-r--  1 student student 469971 May 14 21:06 Task_15.pdf
student@student:~/25SUB5936_U21/Assignments/Day32$ vi buggy.c
student@student:~/25SUB5936_U21/Assignments/Day32$ gcc -g buggy.c -o buggy
student@student:~/25SUB5936_U21/Assignments/Day32$ gdb ./buggy
GNU gdb (Ubuntu 15.1-1ubuntu1~24.04.1) 15.1
Copyright (C) 2024 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
  <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./buggy...
(gdb) break 6
Breakpoint 1 at 0x1155: file buggy.c, line 8.
(gdb) |
```

```
student@student: ~/25SUB5936_U21/Assignments/Day32
Find the GDB manual and other documentation resources online at:
  <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./buggy...
(gdb) break 6
Breakpoint 1 at 0x1155: file buggy.c, line 8.
(gdb) run
Starting program: /home/student/25SUB5936_U21/Assignments/Day32/buggy

This GDB supports auto-downloading debuginfo from the following URLs:
  <https://debuginfod.ubuntu.com>
Enable debuginfod for this session? (y or [n]) y
Debuginfod has been enabled.
To make this setting permanent, add 'set debuginfod enabled on' to .gdbinit.
Downloading separate debug info for system-supplied DSO at 0x7ffff7fc3000
Download failed: Timer expired. Continuing without separate debug info for system-supplied DSO at 0x7ffff7fc3000.
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 1, main () at buggy.c:8
      int i = 1;
(gdb) print i
$1 = 32767
(gdb) next
9      while (i > 5) //Logical Error
(gdb) next
15     printf("Loop finished. Value of i is: %d\n", i);
(gdb) |
```